

OUTPUT

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IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Madhu Bavana/AppData/Local/Programs/Python/Python314/python_code...py
=====
PROBLEM 1: HOSPITAL DOCTOR SHIFT ASSIGNMENT
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Valid Shift Assignment:
D1 -> Morning
D2 -> Afternoon
D3 -> Night

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PROBLEM 2: ROBOT GRID NAVIGATION
=====

Shortest Path:
[(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
Total Cost = 8

=====
PROBLEM 3: AUTONOMOUS RESCUE ROBOT
=====

Least Cost Path:
[(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 3), (2, 4), (3, 4), (4, 4)]
Total Cost = 8

=====
ALL THREE AI PROBLEMS EXECUTED SUCCESSFULLY
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>>>
```